IIIrd Test

INSTITUTE OF ENGINEERING & TECHNOLOGY

BE (II) EI – 2023 : Analog Electronics: 3EIRC4

MM: 20

1. (a) What are the advantages and disadvantages of feedback amplifiers with respect to the amplifiers without the feedback; also list and draw block diagrams for the various types of feedback amplifiers. Derive an expression for the gain with feedback for voltage series feedback amplifier — 6

   (b) Using relevant mathematics determine an expression for gain stability of a feedback amplifier. For a feedback amplifier of A=1000; $\frac{dA_f}{A_f} = 0.01\%$; dA=10. Find the transfer function of the feedback network. — 4

2. (a) Elaborate on the basic elements of an oscillator. Explain Barkhausen criterion. — 4

   (b) A phase shift oscillator, operating at a frequency of 1KHz has a feedback network having all the capacitors of same capacitance, 10micro farads each. If the resistance values of all the resistors is also same, determine its value. — 2

   (c) What is understood by class of amplifiers? Why can't a conventional voltage or a current amplifier work as power amplifier to deliver high power. Also, compare power amplifier with respect to the small signal amplifiers. — 4